Anjali Roy Rathi

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Summary

MSc Statistical Data Science graduate with strong expertise in machine learning, deep learning, and statistical modelling. Experienced in building predictive models, analysing real-world datasets, and deploying end-to-end ML systems. Skilled in Python-based development and API deployment, with a strong interest in solving business problems using data-driven approaches and optimisation techniques.

Education

University of Exeter, UK — Jan 2025 – Jan 2026
MSc Statistical Data Science (Merit)
Relevant Coursework: Machine Learning, Statistical Modelling, Data Mining

SZABIST, Pakistan — Sep 2020 – Aug 2024
BSc Software Engineering (CGPA: 3.53)
Relevant Coursework: Data Structures, Algorithms, Database Systems

Skills

Programming: Python, SQL
Machine Learning: Logistic Regression, Random Forest, XGBoost, SVM
Deep Learning: Neural Networks, CNNs, UNet, RCAN, TensorFlow, Keras
Libraries: Scikit-learn, NumPy, Pandas, SciPy
Concepts: Feature Engineering, Model Evaluation, Optimisation, Cross-validation
Tools: Jupyter Notebook, Git, FastAPI

Projects

Customer Churn Prediction System

- Developed and compared multiple ML models (Logistic Regression, Random Forest, XGBoost) for churn prediction
- Performed data preprocessing, feature engineering, and handling of class imbalance
- Improved model performance using threshold tuning and evaluation metrics such as ROC-AUC and F1-score
- Built and deployed a FastAPI-based REST API to enable real-time prediction

MRI Super Resolution using Deep Learning

- Designed deep learning architectures (UNet, RCAN) to enhance resolution of medical images
- Trained and evaluated models using image quality metrics such as PSNR, SSIM, and RMSE
- Applied regularisation techniques (dropout) to reduce overfitting and improve generalisation

Machine Learning Coursework Projects

- Built regression and classification models on structured datasets using Scikit-learn
- Applied feature engineering, data preprocessing, and cross-validation techniques
- Conducted model comparison and hyperparameter tuning to optimise performance
- Developed a simple FastAPI endpoint to serve trained models for inference

Additional Information

Mathematical Skills: Strong foundation in statistics, probability, and optimisation
Interests: Machine Learning, Deep Learning, NLP